Differential Equations Workbook Solutions

First order equations

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CLASSIFYING DIFFERENTIAL EQUATIONS

■ 1. Identify the order and linearity of the differential equation.

$$5y''' + 4xy = x^2$$

Solution:

The equation $5y''' + 4xy = x^2$ contains a third derivative, so it's a third order equation. We can see that the equation is linear, because the equation does not contain functions of y other than y itself and its derivatives.

■ 2. Say whether or not the linear equation is homogeneous.

$$y'' - x = 3y$$

Solution:

We can rewrite the equation as $y'' - 3y = x$, which allows us to see that the equation is linear and non-homogeneous.

■ 3. Identify the order and linearity of each differential equation.

$$x^3 + 3x \sin x = y'$$

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$$x\frac{d^4y}{dx^4} - x\ln x = 0$$

$$y'' - 3y' + xy^2 = x$$

Solution:

The equation $x^3 + 3x\sin x = y'$ contains a first derivative, so it's a first order equation. We can rewrite it as $y' = x^3 + 3x\sin x$, which allows us to see that the equation is linear.

The equation

$$x\frac{d^4y}{dx^4} - x\ln x = 0$$

contains a fourth derivative, so it's a fourth order equation. We can rewrite it as $d^4y/dx^4 = \ln x$, which allows us to see that the equation is linear.

The equation $y'' - 3y' + xy^2 = x$ contains a second derivative, so it's a second order equation. Because the equation contains y^2, it's non-linear.

■ 4. Determine whether each linear equation is homogeneous or non-homogeneous.

$$y' - 3y = 0$$

$$(\sin x)y' = 0$$

$$\ln x - 3y'' = y$$

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Solution:

Since $q(x) = 0$, the equation $y' - 3y = 0$ is homogeneous.

Since $q(x) = 0$, the equation $(\sin x)y' = 0$ is homogeneous.

We can rewrite $\ln x - 3y'' = y$ as $3y'' + y = \ln x$, which allows us to see that the equation is linear and non-homogeneous.

■ 5. Identify the order and linearity of each differential equation, then say whether or not each linear equation is homogeneous.

$$5y' - y''' - \ln y = x$$

$$y'' = 4y^2$$

$$y' = xy$$

Solution:

The equation $5y' - y''' - \ln y = x$ contains a third derivative, so it's a third order equation. We can rewrite it as $y''' - 5y' = -x - \ln y$, which allows us to see that the equation is non-linear, because the equation contains $\ln y$.

The equation $y'' = 4y^2$ contains a second derivative, so it's a second order equation. Because the equation contains y^2, it's non-linear.

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The equation $y' = xy$ contains a first derivative, so it's a first order equation. We can rewrite it as $y' - xy = 0$, which allows us to see that the equation is linear and homogeneous.

■ 6. Identify the order and linearity of the differential equation.

$$e^x y''' = e^y x$$

Solution:

The equation $e^x y''' = e^y x$ contains a third derivative, so it's a third order equation. Because the equation contains e^y, it's non-linear.

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LINEAR EQUATIONS

■ 1. Find the solution to the linear differential equation.

$$y' - y \sin x = 0$$

Solution:

The linear differential equation is already in standard form, so we can use $P(x) = -\sin x$ to find the integrating factor.

$$\mu(x) = e^{\int -\sin x \, dx}$$

$$\mu(x) = e^{\cos x}$$

Multiply through the differential equation by the integrating factor.

$$e^{\cos x}\left(\frac{dy}{dx} - y \sin x = 0\right)$$

$$e^{\cos x}\frac{dy}{dx} - ye^{\cos x} \sin x = 0$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(ye^{\cos x}) = 0$$

then integrate and solve for y.

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$$\int \frac{d}{dx}(ye^{\cos x}) = \int 0 \, dx$$

$$ye^{\cos x} = C$$

$$y = Ce^{-\cos x}$$

2. Solve the differential equation.

$$y' \cos x + y \sin x = 1$$

Solution:

Put the linear differential equation in standard form.

$$y' \cos x + y \sin x = 1$$

$$y' + y \tan x = \sec x$$

Use $P(x) = \tan x$ to find the integrating factor.

$$\mu(x) = e^{\int \tan x \, dx}$$

$$\mu(x) = e^{-\ln(\cos x)}$$

$$\mu(x) = \frac{1}{\cos x}$$

Multiply through the differential equation by the integrating factor.

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$$\frac{1}{\cos x}(y' + y\tan x = \sec x)$$

$$\frac{1}{\cos x}y' + y\frac{\tan x}{\cos x} = \frac{\sec x}{\cos x}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}\left(\frac{1}{\cos x}y\right) = \frac{1}{\cos^2 x}$$

then integrate and solve for y.

$$\int \frac{d}{dx}\left(\frac{1}{\cos x}y\right) = \int \frac{1}{\cos^2 x}\,dx$$

$$\frac{1}{\cos x}y = \tan x + C$$

$$y = \cos x(\tan x + C)$$

$$y = \sin x + C\cos x$$

3. Find the solution to the linear differential equation.

$$xy' + y = e^x$$

Solution:

Put the linear differential equation in standard form.

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$$y' + \frac{1}{x}y = \frac{e^x}{x}$$

Use $P(x) = 1/x$ to find the integrating factor.

$$\mu(x) = e^{\int \frac{1}{x}\,dx}$$

$$\mu(x) = e^{\ln x}$$

$$\mu(x) = x$$

Multiply through the differential equation by the integrating factor.

$$x\left(y' + \frac{1}{x}y = \frac{e^x}{x}\right)$$

$$xy' + y = e^x$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(xy) = e^x$$

then integrate and solve for y.

$$xy = e^x + C$$

$$y = \frac{e^x + C}{x}$$

4. Solve the differential equation.

$$xy' = x^3 - 3x^3y$$

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Solution:

Put the linear differential equation in standard form.

$$xy' + 3x^3y = x^3$$

$$y' + 3x^2y = x^2$$

Use $P(x) = 3x^2$ to find the integrating factor.

$$\mu(x) = e^{\int 3x^2\ dx}$$

$$\mu(x) = e^{x^3}$$

Multiply through the differential equation by the integrating factor.

$$e^{x^3}(y' + 3x^2y = x^2)$$

$$e^{x^3}y' + 3x^2ye^{x^3} = x^2e^{x^3}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(e^{x^3}y) = x^2e^{x^3}$$

then integrate, using a substitution with $u = x^3$ and $du = 3x^2\ dx$ to integrate the right side.

$$\int \frac{d}{dx}(e^{x^3}y) = \int x^2e^{x^3}\ dx$$

$$e^{x^3}y = \int \frac{e^u}{3}\ du$$

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$$e^{x^3}y = \frac{1}{3}e^u + C$$

$$e^{x^3}y = \frac{1}{3}e^{x^3} + C$$

Solve for y to find the solution to the linear differential equation.

$$y = \frac{1}{3} + Ce^{-x^3}$$

■ 5. Find the solution to the linear differential equation.

$$xy' - 3y = x^2 - x + 1$$

Solution:

Put the linear differential equation in standard form.

$$xy' - 3y = x^2 - x + 1$$

$$y' - \frac{3}{x}y = x - 1 + \frac{1}{x}$$

Now we can use $P(x) = -3/x$ to find the integrating factor.

$$\mu(x) = e^{\int -\frac{3}{x}\,dx}$$

$$\mu(x) = e^{-3\ln x}$$

$$\mu(x) = x^{-3}$$

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Multiply through the differential equation by the integrating factor.

$$x^{-3}\left(y' - \frac{3}{x}y = x - 1 + \frac{1}{x}\right)$$

$$x^{-3}y' - 3x^{-4}y = x^{-2} - x^{-3} + x^{-4}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(x^{-3}y) = x^{-2} - x^{-3} + x^{-4}$$

then integrate and solve for y.

$$\int \frac{d}{dx}(x^{-3}y) = \int x^{-2} - x^{-3} + x^{-4}\ dx$$

$$x^{-3}y = -x^{-1} + \frac{1}{2}x^{-2} - \frac{1}{3}x^{-3} + C$$

$$x^{-3}y = -\frac{1}{x} + \frac{1}{2x^2} - \frac{1}{3x^3} + C$$

$$y = Cx^3 - x^2 + \frac{1}{2}x - \frac{1}{3}$$

6. Solve the differential equation.

$$xy' - y = \sqrt{x}$$

Solution:

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Put the linear differential equation in standard form.

$$y' - \frac{y}{x} = \frac{1}{\sqrt{x}}$$

Now we can use $P(x) = -1/x$ to find the integrating factor.

$$\mu(x) = e^{\int -\frac{1}{x}\, dx}$$

$$\mu(x) = e^{-\ln x}$$

$$\mu(x) = -\frac{1}{x}$$

Multiply through the differential equation by the integrating factor.

$$-\frac{1}{x}\left(y' - \frac{y}{x} = \frac{1}{\sqrt{x}}\right)$$

$$-\frac{1}{x}y' + \frac{y}{x^2} = -\frac{1}{x\sqrt{x}}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}\left(-\frac{y}{x}\right) = -\frac{1}{x\sqrt{x}}$$

then integrate and solve for y.

$$\int \frac{d}{dx}\left(-\frac{y}{x}\right) = \int -\frac{1}{x\sqrt{x}}\, dx$$

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$$-\frac{y}{x} = \frac{2}{\sqrt{x}} + C$$

$$\frac{y}{x} = -\frac{2}{\sqrt{x}} + C$$

$$y = -2\sqrt{x} + Cx$$

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INITIAL VALUE PROBLEMS

■ 1. Solve the initial value problem if $y(0) = 0$.

$$5y' - 10xy = 25x$$

Solution:

Put the linear differential equation in standard form.

$$y' - 2xy = 5x$$

Now use $P(x) = -2x$ to find the integrating factor.

$$\mu(x) = e^{\int -2x\ dx}$$

$$\mu(x) = e^{-x^2}$$

Multiply through the differential equation by the integrating factor.

$$e^{-x^2}\left(\frac{dy}{dx} - 2xy = 5x\right)$$

$$e^{-x^2}\frac{dy}{dx} - 2xye^{-x^2} = 5xe^{-x^2}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(e^{-x^2}y) = 5xe^{-x^2}$$

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then integrate, using a substitution with $u = -x^2$ and $du = -2x\ dx$ to integrate the right side.

$$\int \frac{d}{dx}(e^{-x^2}y) = \int 5xe^{-x^2}\ dx$$

$$e^{-x^2}y = \int -\frac{5}{2}e^{u}\ du$$

$$e^{-x^2}y = -\frac{5}{2}e^{u} + C$$

$$e^{-x^2}y = -\frac{5}{2}e^{-x^2} + C$$

$$y = -\frac{5}{2} + Ce^{x^2}$$

Once we have this general solution, we recognize from the initial condition $y(0) = 0$ that $x = 0$ and $y = 0$, so we'll plug these values into the general solution to solve for C.

$$0 = -\frac{5}{2} + Ce^{(0)^2}$$

$$0 = -\frac{5}{2} + C$$

$$C = \frac{5}{2}$$

So the solution to the differential equation is

$$y = -\frac{5}{2} + \frac{5}{2}e^{x^2}$$

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■ 2. Solve the initial value problem if $y(1) = -3$.

$$x\frac{dy}{dx} + 2y = 6x^2$$

Solution:

Put the linear differential equation in standard form.

$$x\frac{dy}{dx} + 2y = 6x^2$$

$$\frac{dy}{dx} + \frac{2}{x}y = 6x$$

Now we can use $P(x) = 2/x$ to find the integrating factor.

$$\mu(x) = e^{\int \frac{2}{x}\,dx}$$

$$\mu(x) = e^{2\ln x}$$

$$\mu(x) = x^2$$

Multiply through the differential equation by the integrating factor.

$$x^2\left(\frac{dy}{dx} + \frac{2}{x}y = 6x\right)$$

$$x^2\frac{dy}{dx} + 2xy = 6x^3$$

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Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(x^2y) = 6x^3$$

then integrate.

$$\int \frac{d}{dx}(x^2y) = \int 6x^3 \, dx$$

$$x^2y = \frac{3}{2}x^4 + C$$

$$y = \frac{3}{2}x^2 + Cx^{-2}$$

Once we have this general solution, we recognize from the initial condition $y(1) = -3$ that $x = 1$ and $y = -3$, so we'll plug these values into the general solution to solve for C.

$$-3 = \frac{3}{2}(1)^2 + C(1)^{-2}$$

$$-3 = \frac{3}{2} + C$$

$$C = -\frac{9}{2}$$

So the solution to the differential equation is

$$y = \frac{3}{2}x^2 - \frac{9}{2}x^{-2}$$

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■ 3. Solve the initial value problem if $y(0) = 1$.

$$y' - y = 2\sin(3x)$$

Solution:

The linear differential equation is already in standard form, so we can use $P(x) = -1$ to find the integrating factor.

$$\mu(x) = e^{\int -1\, dx}$$

$$\mu(x) = e^{-x}$$

Multiply through the differential equation by the integrating factor.

$$e^{-x}\left(\frac{dy}{dx} - y = 2\sin(3x)\right)$$

$$e^{-x}\frac{dy}{dx} - ye^{-x} = 2\sin(3x)e^{-x}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(e^{-x}y) = 2\sin(3x)e^{-x}$$

then integrate, using integration by parts to integrate the right side.

$$\int \frac{d}{dx}(e^{-x}y) = \int 2\sin(3x)e^{-x}\, dx$$

$$e^{-x}y = -\frac{1}{5}e^{-x}(\sin(3x) + 3\cos(3x)) + C$$

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$$y = -\frac{1}{5}(\sin(3x) + 3\cos(3x)) + Ce^x$$

Once we have this general solution, we recognize from the initial condition $y(0) = 1$ that $x = 0$ and $y = 1$, so we'll plug these values into the general solution to solve for C.

$$1 = -\frac{1}{5}(\sin(3(0)) + 3\cos(3(0))) + Ce^0$$

$$1 = -\frac{3}{5} + C$$

$$C = \frac{8}{5}$$

So the solution to the differential equation is

$$y = -\frac{1}{5}(\sin(3x) + 3\cos(3x)) + \frac{8}{5}e^x$$

■ 4. A function $y(x)$ is a solution of the differential equation. Suppose that $y(1) = 1$ and $y(3) = 3$. Find the constant k and the solution $y(x)$.

$$y' - \frac{ky}{x} = 0$$

Solution:

The linear differential equation is already in standard form, so we can use $P(x) = -k/x$ to find the integrating factor.

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$$\mu(x) = e^{\int -\frac{k}{x}\,dx}$$

$$\mu(x) = e^{-k\ln x}$$

$$\mu(x) = x^{-k}$$

Multiply through the differential equation by the integrating factor.

$$x^{-k}\left(\frac{dy}{dx} - \frac{ky}{x} = 0\right)$$

$$x^{-k}\frac{dy}{dx} - \frac{ky}{x}x^{-k} = 0$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(x^{-k}y) = 0$$

then integrate.

$$\int \frac{d}{dx}(x^{-k}y) = \int 0\,dx$$

$$x^{-k}y = C$$

$$y = Cx^k$$

Once we have this general solution, we recognize from the initial condition $y(1) = 1$ that $x = 1$ and $y = 1$ and from the initial condition $y(3) = 3$ that $x = 3$ and $y = 3$, so we'll plug these values into the general solution,

$$1 = C(1)^k$$

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$$3 = C(3)^k$$

and then simplify to find C and k.

$$C = 1$$

$$3 = 1(3)^k$$

$$k = 1$$

So the solution to the differential equation is

$$y = 1x^1$$

$$y = x$$

■ 5. Solve the initial value problem if $y(\ln 2) = 1$.

$$\frac{dy}{dx} - 6y = 2$$

Solution:

The linear differential equation is already in standard form, so we can use $P(x) = -6$ to find the integrating factor.

$$\mu(x) = e^{\int -6\ dx}$$

$$\mu(x) = e^{-6x}$$

Multiply through the differential equation by the integrating factor.

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$$e^{-6x}\left(\frac{dy}{dx} - 6y = 2\right)$$

$$e^{-6x}\frac{dy}{dx} - 6ye^{-6x} = 2e^{-6x}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(e^{-6x}y) = 2e^{-6x}$$

then integrate.

$$\int \frac{d}{dx}(e^{-6x}y) = \int 2e^{-6x}\, dx$$

$$e^{-6x}y = -\frac{1}{3}e^{-6x} + C$$

$$y = -\frac{1}{3} + Ce^{6x}$$

Once we have this general solution, we recognize from the initial condition $y(\ln 2) = 1$ that $x = \ln 2$ and $y = 1$, so we'll plug these values into the general solution to solve for C.

$$1 = -\frac{1}{3} + Ce^{6\ln 2}$$

$$1 = -\frac{1}{3} + 2^6 C$$

$$C = \frac{1}{48}$$

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So the particular solution to the differential equation is

$$y = -\frac{1}{3} + \frac{1}{48}e^{6x}$$

■ 6. A function $y(x)$ is a solution of $y' + x^k y = 0$. Suppose that $y(0) = 1$ and $y(1) = e^{-5}$. Find the constant k and the solution $y(x)$.

Solution:

The linear differential equation is already in standard form, so we can use $P(x) = x^k$ to find the integrating factor.

$$\mu(x) = e^{\int x^k \, dx}$$

$$\mu(x) = e^{\frac{x^{k+1}}{k+1}}$$

Multiply through the differential equation by the integrating factor.

$$e^{\frac{x^{k+1}}{k+1}}\left(\frac{dy}{dx} + x^k y = 0\right)$$

$$e^{\frac{x^{k+1}}{k+1}}\frac{dy}{dx} + x^k y e^{\frac{x^{k+1}}{k+1}} = 0$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(e^{\frac{x^{k+1}}{k+1}}y) = 0$$

then integrate.

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$$\int \frac{d}{dx}(e^{\frac{x^{k+1}}{k+1}}y) = \int 0 \, dx$$

$$e^{\frac{x^{k+1}}{k+1}}y = C$$

$$y = Ce^{-\frac{x^{k+1}}{k+1}}$$

Once we have this general solution, we recognize from the initial condition $y(0) = 1$ that $x = 0$ and $y = 1$, and from the initial condition $y(1) = e^{-5}$ that $x = 1$ and $y = e^{-5}$, so we'll plug these values into the general solution,

$$1 = Ce^{-\frac{0^{k+1}}{k+1}}$$

$$e^{-5} = Ce^{-\frac{1^{k+1}}{k+1}}$$

and then simplify these to solve for C and k.

$$C = 1$$

$$e^{-5} = e^{-\frac{1}{k+1}}$$

$$5(k+1) = 1$$

$$k + 1 = \frac{1}{5}$$

$$k = -\frac{4}{5}$$

So the solution to the differential equation is

$$y = e^{-5x^{\frac{1}{5}}}$$

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SEPARABLE EQUATIONS

■ 1. Find the solution to the separable differential equation.

$$\frac{dy}{dx} = \frac{3x^2}{y}$$

Solution:

Separate variables, moving y to the left and x to the right.

$$dy = \frac{3x^2}{y}\,dx$$

$$y\,dy = 3x^2\,dx$$

Integrate both sides, adding the constant of integration C to the right side.

$$\int y\,dy = \int 3x^2\,dx$$

$$\frac{1}{2}y^2 = x^3 + C$$

Solve for y to find the solution to the separable differential equation.

$$y^2 = 2(x^3 + C)$$

$$y = \pm\sqrt{2(x^3 + C)}$$

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■ 2. Find the solution to the separable differential equation.

$$\frac{dy}{dx} = \frac{x+2}{2y}$$

Solution:

Separate variables, moving y to the left and x to the right.

$$dy = \frac{x+2}{2y}\,dx$$

$$2y\,dy = (x+2)\,dx$$

Integrate both sides, adding the constant of integration C to the right side.

$$\int 2y\,dy = \int x+2\,dx$$

$$y^2 = \frac{1}{2}x^2 + 2x + C$$

Solve for y to find the solution to the separable differential equation.

$$y = \pm\sqrt{\frac{1}{2}x^2 + 2x + C}$$

■ 3. Find the solution to the separable differential equation.

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$$\frac{dy}{dx} = \frac{4x^3 - 1}{x(y^2 - 1)}$$

Solution:

Separate variables, moving y to the left and x to the right.

$$dy = \frac{4x^3 - 1}{x(y^2 - 1)}\, dx$$

$$(y^2 - 1)\, dy = \frac{4x^3 - 1}{x}\, dx$$

$$(y^2 - 1)\, dy = \left(\frac{4x^3}{x} - \frac{1}{x}\right)\, dx$$

$$(y^2 - 1)\, dy = \left(4x^2 - \frac{1}{x}\right)\, dx$$

Integrate both sides, adding the constant of integration C to the right side.

$$\int y^2 - 1\, dy = \int 4x^2 - \frac{1}{x}\, dx$$

$$\frac{1}{3}y^3 - y = \frac{4}{3}x^3 - \ln|x| + C$$

Solve for y to find the solution to the separable differential equation.

$$y^3 - 3y = 4x^3 - 3\ln|x| + 3C$$

$$y^3 - 3y = 4x^3 - 3\ln|x| + C$$

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We can't solve this equation explicitly for y, so we'll leave this as the implicitly-defined general solution.

■ 4. Find the solution to the separable differential equation.

$$\frac{dy}{dx} = \frac{x^2\sqrt{x^3 - 3}}{y^2}$$

Solution:

Start by separating variables, putting y terms on the left and x terms on the right.

$$dy = \frac{x^2\sqrt{x^3 - 3}}{y^2}\, dx$$

$$y^2\, dy = x^2\sqrt{x^3 - 3}\, dx$$

Now integrate both sides, using a substitution with $u = x^3 - 3$ and $du/3 = x^2\, dx$, and adding the constant of integration C to the right side.

$$\int y^2\, dy = \int \sqrt{u}\left(\frac{du}{3}\right)$$

$$\frac{1}{3}y^3 = \frac{2}{9}u^{\frac{3}{2}} + C$$

Back-substitute for $u = x^3 - 3$.

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$$\frac{1}{3}y^3 = \frac{2}{9}(x^3 - 3)^{\frac{3}{2}} + C$$

Solve for y to get the explicit the solution to the separable differential equation.

$$y^3 = \frac{2}{3}(x^3 - 3)^{\frac{3}{2}} + C$$

$$y = \left(\frac{2}{3}(x^3 - 3)^{\frac{3}{2}} + C\right)^{\frac{1}{3}}$$

■ 5. Find the solution to the separable differential equation.

$$y' + \sin(x - y) = \sin(x + y)$$

Solution:

Separate variables, moving y to the left and x to the right.

$$y' = \sin(x + y) - \sin(x - y)$$

$$y' = 2\cos x \sin y$$

$$dy = 2\cos x \sin y \; dx$$

$$\frac{1}{\sin y} \; dy = 2\cos x \; dx$$

Integrate both sides, adding the constant of integration C to the right side.

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$$\int \frac{1}{\sin y}\, dy = \int 2\cos x\ dx$$

$$\ln\left|\tan\frac{y}{2}\right| = 2\sin x + C$$

■ 6. Solve the initial value problem if $y(0) = -1$.

$$(xy^2 + x) + (x^2y - y)y' = 0$$

Solution:

Separate variables, moving y to the left and x to the right.

$$(xy^2 + x) + (x^2y - y)y' = 0$$

$$(x^2y - y)y' = -(xy^2 + x)$$

$$(x^2y - y)\ dy = -(xy^2 + x)\ dx$$

$$(x^2 - 1)y\ dy = -(y^2 + 1)x\ dx$$

$$\frac{y}{y^2 + 1}\ dy = -\frac{x}{x^2 - 1}\ dx$$

Integrate both sides, using substitutions on both sides, and adding the constant of integration C to the right side.

$$\int \frac{y}{y^2 + 1}\ dy = \int -\frac{x}{x^2 - 1}\ dx$$

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$$\frac{1}{2}\ln|y^2+1| = -\frac{1}{2}\ln|x^2-1| + C$$

Once we have this general solution, we recognize from the initial condition $y(0) = -1$ that $x = 0$ and $y = -1$, so we'll plug these values into the general solution,

$$\frac{1}{2}\ln(2) = -\frac{1}{2}\ln(1) + C$$

$$\frac{1}{2}\ln(2) = C$$

So the solution to the differential equation is

$$\frac{1}{2}\ln(y^2+1) = -\frac{1}{2}\ln|x^2-1| + \frac{1}{2}\ln(2)$$

$$\ln(y^2+1) = -\ln|x^2-1| + \ln(2)$$

$$\ln(y^2+1) = \ln\frac{2}{|x^2-1|}$$

$$y^2+1 = \frac{2}{|x^2-1|}$$

$$y^2 = \frac{2}{|x^2-1|} - 1$$

$$y = \pm\sqrt{\frac{2}{|x^2-1|} - 1}$$

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SUBSTITUTIONS

■ 1. Use a substitution to solve the separable differential equation.

$$y' = \sin(x + y)$$

Solution:

If we choose the substitution $u = ax + by$, then we can set $u = x + y$, $u' = 1 + y'$, and $y' = u' - 1$. Then we'll substitute $y' = u' - 1$ into the left side of the original differential equation, and $u = x + y$ into the right side of the original differential equation, and we'll get

$$u' - 1 = \sin u$$

Now the equation is separable, so we'll separate variables,

$$\frac{du}{dx} = \sin u + 1$$

$$du = (\sin u + 1)\, dx$$

$$\frac{1}{\sin u + 1}\, du = dx$$

and then integrate both sides.

$$\int \frac{1}{\sin u + 1}\, du = \int dx$$

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$$\frac{2}{\cot\frac{u}{2}+1}=x+C$$

$$\frac{2}{x+C}=\cot\frac{u}{2}+1$$

$$\frac{2}{x+C}-1=\cot\frac{u}{2}$$

We'll back-substitute using $u=x+y$, leaving the equation solution implicitly defined in x and y.

$$\frac{2}{x+C}-1=\cot\left(\frac{x+y}{2}\right)$$

2. Use a substitution to solve the separable differential equation.

$$y'=(x+y-5)^2$$

Solution:

If we choose the substitution $u=ax+by$, then we can set up $u=x+y$, $u'=1+y'$, and $y'=u'-1$. Then we'll substitute $y'=u'-1$ into the left side of the original differential equation, and $u=x+y$ into the right side of the original differential equation, and we'll get

$$u'-1=(u-5)^2$$

Now the equation is separable, so we'll separate variables,

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$$\frac{du}{dx} = (u-5)^2 + 1$$

$$du = ((u-5)^2 + 1)\, dx$$

$$\frac{1}{(u-5)^2+1}\, du = dx$$

and then integrate both sides.

$$\int \frac{1}{(u-5)^2+1}\, du = \int dx$$

$$-\tan^{-1}(5-u) = x + C$$

$$\tan^{-1}(5-u) = -x + C$$

$$\tan(\tan^{-1}(5-u)) = \tan(-x + C)$$

$$5 - u = \tan(-x + C)$$

$$u = 5 - \tan(-x + C)$$

We'll back-substitute using $u = x + y$ and then solve for y to find the solution to the differential equation.

$$x + y = 5 - \tan(-x + C)$$

$$y = 5 - \tan(-x + C) - x$$

■ 3. Use a substitution to solve the initial value problem, if $y(0) = 0$.

$$y' = e^{5y-x}$$

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Solution:

If we choose the substitution $u = ax + by$, then we can set $u = 5y - x$, $u' = 5y' - 1$, and $y' = (u' + 1)/5$. Then we'll substitute $y' = (u' + 1)/5$ into the left side of the original differential equation, and $u = 5y - x$ into the right side of the original differential equation, and we'll get

$$\frac{u' + 1}{5} = e^u$$

$$u' + 1 = 5e^u$$

$$u' = 5e^u - 1$$

Now the equation is separable, so we'll separate variables,

$$\frac{du}{dx} = 5e^u - 1$$

$$du = (5e^u - 1)\, dx$$

$$\frac{1}{5e^u - 1}\, du = dx$$

and then integrate both sides.

$$\int \frac{1}{5e^u - 1}\, du = \int dx$$

$$\int \frac{e^{-u}}{5 - e^{-u}}\, du = \int dx$$

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$$\ln(5 - e^{-u}) = x + C$$

$$e^{\ln(5-e^{-u})} = e^{x+C}$$

$$5 - e^{-u} = Ce^{x}$$

$$e^{-u} = 5 - Ce^{x}$$

$$-u = \ln(5 - Ce^{x})$$

$$u = -\ln(5 - Ce^{x})$$

We'll back-substitute using $u = 5y - x$ and then solve for y to find the solution to the differential equation.

$$5y - x = -\ln(5 - Ce^{x})$$

$$5y = x - \ln(5 - Ce^{x})$$

$$y = \frac{1}{5}(x - \ln(5 - Ce^{x}))$$

Once we have this general solution, we recognize from the initial condition $y(0) = 0$ that $x = 0$ and $y = 0$, so we'll plug these values into the general solution to find C,

$$0 = \frac{1}{5}(0 - \ln(5 - Ce^{0}))$$

$$0 = \ln(5 - C)$$

$$e^{0} = e^{\ln(5-C)^{-1}}$$

$$1 = \frac{1}{5 - C}$$

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$$5 - C = 1$$

$$C = 4$$

and then the solution to the differential equation is

$$y = \frac{1}{5}(x - \ln(5 - 4e^x))$$

■ 4. Use a substitution to solve the separable differential equation.

$$y' = \frac{1}{6x - 3y}$$

Solution:

If we choose the substitution $u = ax + by$, then we can set $u = 6x - 3y$, $u' = 6 - 3y'$, and $y' = 2 - (1/3)u'$. Then we'll substitute $y' = 2 - (1/3)u'$ into the left side of the original differential equation, and $u = 6x - 3y$ into the right side of the original differential equation, and we'll get

$$2 - \frac{1}{3}u' = \frac{1}{u}$$

$$2 - \frac{1}{u} = \frac{1}{3}u'$$

$$\frac{6u - 3}{u} = u'$$

Now the equation is separable, so we'll separate variables,

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$$\frac{du}{dx} = \frac{6u - 3}{u}$$

$$du = \left(\frac{6u - 3}{u}\right) dx$$

$$\frac{u}{6u - 3}\, du = dx$$

and then integrate both sides.

$$\int \frac{u}{6u - 3}\, du = \int dx$$

$$\frac{1}{3}\int \frac{u}{2u - 1}\, du = \int dx$$

$$\frac{1}{6}\left(\int \frac{1}{2u - 1}\, du + \int du\right) = \int dx$$

$$\frac{1}{6}(\ln(2u - 1) + u) = x + C$$

Simplify to find the solution to the differential equation.

$$\ln(2u - 1) + u = 6x + C$$

$$\ln(12x - 6y - 1) + 6x - 3y = 6x + C$$

$$\ln(12x - 6y - 1) - 3y = C$$

■ 5. Use a substitution to solve the initial value problem, if $y(0) = 8$.

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$$y' + (9x + y - 1)^2 = 0$$

Solution:

If we choose the substitution $u = ax + by$, then we can set $u = 9x + y$, $u' = 9 + y'$, and $y' = u' - 9$. Then we'll substitute $y' = u' - 9$ and $u = 9x + y$ into the left side of the original differential equation, and we'll get

$$u' - 9 + (u - 1)^2 = 0$$

$$u' = 9 - (u - 1)^2$$

Now the equation is separable, so we'll separate variables,

$$\frac{du}{dx} = 9 - (u - 1)^2$$

$$du = (9 - (u - 1)^2)\, dx$$

$$\frac{1}{9 - (u - 1)^2}\, du = dx$$

and then integrate both sides.

$$\int \frac{1}{9 - (u - 1)^2}\, du = \int dx$$

$$\int \frac{du}{9 - (u^2 - 2u + 1)} = \int dx$$

$$\int \frac{du}{9 - u^2 + 2u - 1} = \int dx$$

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$$\int \frac{du}{8+2u-u^2} = \int dx$$

$$-\int \frac{du}{u^2-2u-8} = \int dx$$

$$-\int \frac{du}{(u+2)(u-4)} = \int dx$$

Use partial fractions to rewrite the integral on the left.

$$\frac{1}{6}\int \frac{1}{u+2} - \frac{1}{u-4} = \int dx$$

$$\frac{1}{6}(\ln(u+2) - \ln(u-4)) = x + C$$

$$\ln\left(\frac{u+2}{u-4}\right) = 6x + C$$

Raise both sides to the base e, then solve for u.

$$\frac{u+2}{u-4} = Ce^{6x}$$

$$u+2 = (u-4)Ce^{6x}$$

$$u+2 = uCe^{6x} - 4Ce^{6x}$$

$$u - uCe^{6x} = -2 - 4Ce^{6x}$$

$$u(1 - Ce^{6x}) = -2 - 4Ce^{6x}$$

$$u = \frac{-2-4Ce^{6x}}{1-Ce^{6x}}$$

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We'll back-substitute using $u = 9x + y$ and then solve for y to find the solution to the differential equation.

$$9x + y = \frac{-2 - 4Ce^{6x}}{1 - Ce^{6x}}$$

$$y = \frac{-2 - 4Ce^{6x}}{1 - Ce^{6x}} - 9x$$

Once we have this general solution, we recognize from the initial condition $y(0) = 8$ that $x = 0$ and $y = 8$, so we'll plug these values into the general solution to find C.

$$8 = \frac{-2 - 4Ce^{6(0)}}{1 - Ce^{6(0)}} - 9(0)$$

$$8 = \frac{-2 - 4C}{1 - C}$$

$$8(1 - C) = -2 - 4C$$

$$8 - 8C = -2 - 4C$$

$$4C = 10$$

$$C = \frac{5}{2}$$

Then the solution is

$$y = \frac{-2 - 4\frac{5}{2}e^{6x}}{1 - \frac{5}{2}e^{6x}} - 9x$$

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$$y = \frac{-2 - 10e^{6x}}{1 - \frac{5}{2}e^{6x}} - 9x$$

$$y = \frac{-4 - 20e^{6x}}{2 - 5e^{6x}} - 9x$$

$$y = \frac{20e^{6x} + 4}{5e^{6x} - 2} - 9x$$

■ 6. Use a substitution to solve the initial value problem, if $y(0) = 2$.

$$y' + 2x + 2y - 2xy = x^2 + y^2$$

Solution:

We can rewrite this equation as

$$y' + 2x + 2y - 2xy = x^2 + y^2$$

$$y' + 2(x + y) = x^2 + 2xy + y^2$$

$$y' + 2(x + y) = (x + y)^2$$

If we choose the substitution $u = ax + by$, then we can set $u = x + y$, $u' = 1 + y'$, and $y' = u' - 1$. Then we'll substitute $y' = u' - 1$ into the left side of the original differential equation, and $u = x + y$ into the right side of the original differential equation, and we'll get

$$u' - 1 + 2u = u^2$$

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$$u' = u^2 - 2u + 1$$

$$u' = (u-1)^2$$

Now the equation is separable, so we'll separate variables,

$$\frac{du}{dx} = (u-1)^2$$

$$du = (u-1)^2\, dx$$

$$\frac{1}{(u-1)^2}\, du = dx$$

and then integrate both sides.

$$\int \frac{1}{(u-1)^2}\, du = \int dx$$

$$\frac{1}{1-u} = x + C$$

$$\frac{1}{x+C} = 1 - u$$

$$u = 1 - \frac{1}{x+C}$$

We'll back-substitute using $u = x + y$ and then solve for y to find the solution to the differential equation.

$$x + y = 1 - \frac{1}{x+C}$$

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$$y = 1 - \frac{1}{x + C} - x$$

Once we have this general solution, we recognize from the initial condition $y(0) = 2$ that $x = 0$ and $y = 2$, so we'll plug these values into the general solution to find C.

$$2 = 1 - \frac{1}{0 + C} - 0$$

$$2 = 1 - \frac{1}{C}$$

$$1 = -\frac{1}{C}$$

$$C = -1$$

Then the solution is

$$y = 1 - \frac{1}{x - 1} - x$$

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BERNOULLI EQUATIONS

■ 1. Find the solution to the Bernoulli differential equation.

$$(y^4 + x^4y)\,dx - 3x^5\,dy = 0$$

Solution:

Rewrite the equation in standard form.

$$y^4 + x^4y = 3x^5\frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{y^4 + x^4y}{3x^5}$$

Split the fraction so that we can separate terms.

$$\frac{dy}{dx} = \frac{y^4}{3x^5} + \frac{x^4y}{3x^5}$$

$$\frac{dy}{dx} - \frac{x^4y}{3x^5} = \frac{y^4}{3x^5}$$

$$\frac{dy}{dx} - \frac{x^4}{3x^5}y = \frac{1}{3x^5}y^4$$

$$y' - \frac{1}{3x}y = \frac{1}{3x^5}y^4$$

Divide through by $y^n = y^4$.

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$$\frac{y'}{y^4} - \frac{y}{3xy^4} = \frac{y^4}{3x^5y^4}$$

$$y'y^{-4} - \frac{1}{3x}y^{-3} = \frac{1}{3x^5}$$

Our substitution is $v = y^{-3}$, so we'll differentiate to get

$$v' = -3y^{-4}y'$$

and then solve this for $y^{-4}y'$.

$$y^{-4}y' = -\frac{1}{3}v'$$

Now we can make substitutions into the Bernoulli equation.

$$-\frac{1}{3}v' - \frac{1}{3x}v = \frac{1}{3x^5}$$

$$v' + \frac{1}{x}v = -\frac{1}{x^5}$$

To find the solution to the linear equation, we'll find the integrating factor,

$$I(x) = e^{\int \frac{1}{x}\,dx}$$

$$I(x) = e^{\ln x}$$

$$I(x) = x$$

and then multiply through the linear equation by $I(x)$.

$$xv' + \frac{1}{x}xv = -\frac{1}{x^5}x$$

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$$xv' + v = -\frac{1}{x^4}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(vx) = -\frac{1}{x^4}$$

then integrate both sides.

$$\int \frac{d}{dx}(vx) = \int -\frac{1}{x^4}\, dx$$

$$vx = \frac{1}{3x^3} + C$$

Solve for v.

$$v = \frac{1}{3x^4} + \frac{C}{x}$$

Use $v = y^{-3}$ to back-substitute for v,

$$y^{-3} = \frac{1 + 3x^3C}{3x^4}$$

then solve for y to find the solution.

$$\frac{1}{y^3} = \frac{1 + 3x^3C}{3x^4}$$

$$y^3 = \frac{3x^4}{1 + 3x^3C}$$

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$$y = \sqrt[3]{\frac{3x^4}{1 + 3x^3C}}$$

2. Find the solution to the Bernoulli differential equation.

$$xy' - 3y = xy^{\frac{5}{3}}$$

Solution:

Rewrite the equation in standard form.

$$y' - \frac{3y}{x} = y^{\frac{5}{3}}$$

Divide through by $y^n = y^{\frac{5}{3}}$.

$$\frac{y'}{y^{\frac{5}{3}}} - \frac{3y}{xy^{\frac{5}{3}}} = \frac{y^{\frac{5}{3}}}{y^{\frac{5}{3}}}$$

$$y'y^{-\frac{5}{3}} - \frac{3}{x}y^{-\frac{2}{3}} = 1$$

Our substitution is $v = y^{-\frac{2}{3}}$, so we'll differentiate to get

$$v' = -\frac{2}{3}y^{-\frac{5}{3}}y'$$

and then solve this for $y^{-\frac{5}{3}}y'$.

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$$y^{-\frac{5}{3}}y' = -\frac{3}{2}v'$$

Now we can make substitutions into the Bernoulli equation.

$$-\frac{3}{2}v' - \frac{3}{x}v = 1$$

$$v' + \frac{2}{x}v = -\frac{2}{3}$$

To find the solution to the linear equation, we'll find the integrating factor,

$$I(x) = e^{\int \frac{2}{x}\,dx}$$

$$I(x) = e^{2\ln x}$$

$$I(x) = e^{\ln x^2}$$

$$I(x) = x^2$$

and then multiply through the linear equation by $I(x)$.

$$v'(x^2) + \frac{2}{x}v(x^2) = -\frac{2}{3}(x^2)$$

$$x^2v' + 2xv = -\frac{2}{3}x^2$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(x^2v) = -\frac{2}{3}x^2$$

then integrate both sides.

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$$\int \frac{d}{dx}(x^2 v)\, dx = \int -\frac{2}{3}x^2\, dx$$

$$x^2 v = -\frac{2}{9}x^3 + C$$

Solve for v.

$$v = -\frac{2}{9}x + \frac{1}{x^2}C$$

Use $v = y^{-\frac{2}{3}}$ to back-substitute for v,

$$y^{-\frac{2}{3}} = -\frac{2}{9}x + \frac{1}{x^2}C$$

then solve for y.

$$\frac{1}{y^{\frac{2}{3}}} = -\frac{2}{9}x + \frac{1}{x^2}C$$

$$y^{\frac{2}{3}} = \frac{1}{-\frac{2}{9}x + \frac{1}{x^2}C}$$

$$y = \frac{1}{\sqrt{\left(-\frac{2}{9}x + \frac{1}{x^2}C\right)^3}}$$

Simplify and rewrite the solution.

$$y = \frac{1}{\sqrt{\left(\frac{1}{x^2}C - \frac{2}{9}x\right)^3}}$$

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$$y = \frac{1}{\sqrt{\left(\frac{9C}{9x^2} - \frac{2x^3}{9x^2}\right)^3}}$$

$$y = \frac{1}{\left(\frac{9C - 2x^3}{9x^2}\right)^{\frac{3}{2}}}$$

$$y = \frac{1}{\frac{(9C - 2x^3)^{\frac{3}{2}}}{(9x^2)^{\frac{3}{2}}}}$$

$$y = \frac{1}{\frac{(9C - 2x^3)^{\frac{3}{2}}}{27x^3}}$$

$$y = \frac{27x^3}{(C - 2x^3)^{\frac{3}{2}}}$$

■ 3. Find the solution to the Bernoulli differential equation.

$$y' = y^2 \sin x + y \cot x$$

Solution:

Rewrite the equation in standard form.

$$y' - y \cot x = y^2 \sin x$$

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Divide through by $y^n = y^2$.

$$\frac{y'}{y^2} - \frac{y \cot x}{y^2} = \frac{y^2 \sin x}{y^2}$$

$$y'y^{-2} - (\cot x)y^{-1} = \sin x$$

Our substitution is $v = y^{-1}$, so we'll differentiate to get

$$v' = -y^{-2}y'$$

and then solve this for $y^{-2}y'$.

$$y^{-2}y' = -v'$$

Now we can make substitutions into the Bernoulli equation.

$$-v' - (\cot x)v = \sin x$$

$$v' + (\cot x)v = -\sin x$$

To find the solution to the linear equation, we'll find the integrating factor,

$$I(x) = e^{\int \cot x \, dx}$$

$$I(x) = e^{\ln \sin x}$$

$$I(x) = \sin x$$

and then multiply through the linear equation by $I(x)$.

$$v' \sin x + (\cot x \sin x)v = -\sin x \sin x$$

$$v' \sin x + (\cot x \sin x)v = -\sin^2 x$$

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Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(v \sin x) = -\sin^2 x$$

then integrate both sides.

$$\int \frac{d}{dx}(v \sin x) = \int -\sin^2 x \; dx$$

$$v \sin x = -\frac{1}{2}x + \frac{1}{2}\sin x \cos x + C$$

Solve for v.

$$v = -\frac{1}{2}x \csc x + \frac{1}{2}\cos x + C \csc x$$

Use $v = y^{-1}$ to back-substitute for v,

$$y^{-1} = -\frac{1}{2}x \csc x + \frac{1}{2}\cos x + C \csc x$$

then solve for y.

$$\frac{1}{y} = -\frac{1}{2}x \csc x + \frac{1}{2}\cos x + C \csc x$$

$$y = \frac{1}{-\frac{1}{2}x \csc x + \frac{1}{2}\cos x + C \csc x}$$

■ 4. Find the solution to the Bernoulli differential equation, if $y(1) = 0$.

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$$y' - \frac{y}{x} + \sqrt{y} = 0$$

Solution:

Rewrite the equation in standard form.

$$y' - \frac{y}{x} = -\sqrt{y}$$

Divide through by $y^n = y^{\frac{1}{2}}$.

$$\frac{y'}{y^{\frac{1}{2}}} - \frac{y}{xy^{\frac{1}{2}}} = -\frac{y^{\frac{1}{2}}}{y^{\frac{1}{2}}}$$

$$y'y^{-\frac{1}{2}} - \frac{1}{x}y^{\frac{1}{2}} = -1$$

Our substitution is $v = y^{\frac{1}{2}}$, so we'll differentiate to get

$$v' = \frac{1}{2}y^{-\frac{1}{2}}y'$$

and then solve this for $y^{-\frac{1}{2}}y'$.

$$y^{-\frac{1}{2}}y' = 2v'$$

Now we can make substitutions into the Bernoulli equation.

$$2v' - \frac{1}{x}v = -1$$

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Multiplying through by $1/2$ puts the equation into standard form of a linear differential equation.

$$v' - \frac{1}{2x}v = -\frac{1}{2}$$

To find the solution to the linear equation, we'll find the integrating factor,

$$I(x) = e^{\int -\frac{1}{2x}\,dx}$$

$$I(x) = e^{-\frac{1}{2}\ln x}$$

$$I(x) = \frac{1}{x^{\frac{1}{2}}}$$

and then multiply through the linear equation by $I(x)$.

$$\frac{1}{x^{\frac{1}{2}}}v' - \frac{1}{x^{\frac{1}{2}}}\frac{1}{2x}v = -\frac{1}{2}\frac{1}{x^{\frac{1}{2}}}$$

$$\frac{1}{x^{\frac{1}{2}}}v' - \frac{1}{2x^{\frac{3}{2}}}v = -\frac{1}{2x^{\frac{1}{2}}}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}\left(\frac{1}{x^{\frac{1}{2}}}v\right) = -\frac{1}{2x^{\frac{1}{2}}}$$

then integrate both sides.

$$\int \frac{d}{dx}\left(\frac{1}{x^{\frac{1}{2}}}v\right) = \int -\frac{1}{2x^{\frac{1}{2}}}\,dx$$

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$$\frac{1}{x^{\frac{1}{2}}}v = -x^{\frac{1}{2}} + C$$

Solve for v.

$$v = -x + Cx^{\frac{1}{2}}$$

Use $v = y^{\frac{1}{2}}$ to back-substitute for v,

$$y^{\frac{1}{2}} = -x + Cx^{\frac{1}{2}}$$

then solve for y.

$$y = (Cx^{\frac{1}{2}} - x)^2$$

Once we have this general solution, we recognize from the initial condition $y(1) = 0$ that $x = 1$ and $y = 0$, so we'll plug these values into the general solution to solve for C.

$$0 = (C(1)^{\frac{1}{2}} - 1)^2$$

$$0 = (C - 1)^2$$

$$0 = C - 1$$

$$C = 1$$

Then the solution is

$$y = (x^{\frac{1}{2}} - x)^2$$

■ 5. Find the solution to the Bernoulli differential equation, if $y(1) = 1$.

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$$y' + \frac{2}{x}y = \frac{y^2}{x^2}$$

Solution:

Rewrite the equation in standard form.

$$\frac{y'}{y^2} + \frac{2y}{xy^2} = \frac{y^2}{x^2y^2}$$

$$y'y^{-2} + \frac{2}{x}y^{-1} = \frac{1}{x^2}$$

Our substitution is $v = y^{-1}$, so we'll differentiate to get

$$v' = -y^{-2}y'$$

and then solve this for $y^{-2}y'$.

$$y^{-2}y' = -v'$$

Now we can make substitutions into the Bernoulli equation.

$$y'y^{-2} + \frac{2}{x}y^{-1} = \frac{1}{x^2}$$

$$-v' + \frac{2}{x}v = \frac{1}{x^2}$$

$$v' - \frac{2}{x}v = -\frac{1}{x^2}$$

To find the solution to the linear equation, we'll find the integrating factor,

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$$I(x) = e^{\int -\frac{2}{x}\,dx}$$

$$I(x) = e^{-2\ln x}$$

$$I(x) = e^{\ln x^{-2}}$$

$$I(x) = \frac{1}{x^2}$$

and then multiply through the linear equation by $I(x)$.

$$\frac{1}{x^2}v' - \frac{2}{x}\frac{1}{x^2}v = -\frac{1}{x^2}\frac{1}{x^2}$$

$$\frac{1}{x^2}v' - \frac{2}{x^3}v = -\frac{1}{x^4}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}\left(\frac{1}{x^2}v\right) = -\frac{1}{x^4}$$

then integrate both sides.

$$\int \frac{d}{dx}\left(\frac{1}{x^2}v\right) = \int -\frac{1}{x^4}\,dx$$

$$v\frac{1}{x^2} = \frac{1}{3x^3} + C$$

Solve for v.

$$v = \frac{1}{3x} + Cx^2$$

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Use $v = y^{-1}$ to back-substitute for v,

$$y^{-1} = \frac{1}{3x} + Cx^2$$

then solve for y.

$$\frac{1}{y} = \frac{1}{3x} + Cx^2$$

$$y = \frac{1}{\frac{1}{3x} + Cx^2}$$

Once we have this general solution, we recognize from the initial condition $y(1) = 1$ that $x = 1$ and $y = 1$, so we'll plug these values into the general solution to solve for C.

$$1 = \frac{1}{\frac{1}{3(1)} + C(1)^2}$$

$$1 = \frac{1}{\frac{1}{3} + C}$$

$$1 = \frac{1}{3} + C$$

$$C = \frac{2}{3}$$

Then the solution is

$$y = \frac{1}{\frac{1}{3x} + \frac{2}{3}x^2}$$

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■ 6. Find the solution to the Bernoulli differential equation, if $y(0) = 2$.

$$3y' + 2y = xy^5$$

Solution:

Rewrite the equation in standard form.

$$y' + \frac{2}{3}y = \frac{1}{3}xy^5$$

Divide through by $y^n = y^5$.

$$\frac{y'}{y^5} + \frac{2y}{3y^5} = \frac{xy^5}{3y^5}$$

$$y'y^{-5} + \frac{2}{3}y^{-4} = \frac{x}{3}$$

Our substitution is $v = y^{-4}$, so we'll differentiate to get

$$v' = -4y^{-5}y'$$

and then solve this for $y^{-5}y'$.

$$y^{-5}y' = -\frac{1}{4}v'$$

Now we can make substitutions into the Bernoulli equation.

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$$-\frac{1}{4}v' + \frac{2}{3}v = \frac{x}{3}$$

$$v' - \frac{8}{3}v = -\frac{4x}{3}$$

To find the solution to the linear equation, we'll find the integrating factor,

$$I(x) = e^{\int -\frac{8}{3}\,dx}$$

$$I(x) = e^{-\frac{8}{3}x}$$

and then multiply through the linear equation by $I(x)$.

$$v' - \frac{8}{3}v = -\frac{4x}{3}$$

$$e^{-\frac{8}{3}x}v' - \frac{8}{3}e^{-\frac{8}{3}x}v = -\frac{4x}{3}e^{-\frac{8}{3}x}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(ve^{-\frac{8}{3}x}) = -\frac{4}{3}xe^{-\frac{8}{3}x}$$

then integrate both sides.

$$\int \frac{d}{dx}(ve^{-\frac{8}{3}x}) = \int -\frac{4}{3}xe^{-\frac{8}{3}x}\,dx$$

$$ve^{-\frac{8}{3}x} = \frac{1}{16}e^{-\frac{8}{3}x}(8x+3) + C$$

Solve for v.

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$$v = \frac{1}{16}(8x + 3) + Ce^{\frac{8}{3}x}$$

Use $v = y^{-4}$ to back-substitute for v,

$$y^{-4} = \frac{1}{16}(8x + 3) + Ce^{\frac{8}{3}x}$$

then solve for y.

$$\frac{1}{y^4} = \frac{1}{16}(8x + 3) + Ce^{\frac{8}{3}x}$$

$$y^4 = \frac{1}{\frac{1}{16}(8x + 3) + Ce^{\frac{8}{3}x}}$$

$$y = \frac{1}{\sqrt[4]{\frac{1}{16}(8x + 3) + Ce^{\frac{8}{3}x}}}$$

Once we have this general solution, we recognize from the initial condition $y(0) = 2$ that $x = 0$ and $y = 2$, so we'll plug these values into the general solution to find C.

$$2 = \frac{1}{\sqrt[4]{\frac{1}{16}(8(0) + 3) + Ce^{\frac{8}{3}(0)}}}$$

$$2 = \frac{1}{\sqrt[4]{\frac{3}{16} + C}}$$

$$16 = \frac{1}{\frac{3}{16} + C}$$

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$$\frac{3}{16} + C = \frac{1}{16}$$

$$C = -\frac{1}{8}$$

Then the solution is

$$y = \frac{1}{\sqrt[4]{\frac{1}{16}(8x+3) - \frac{1}{8}e^{\frac{8}{3}x}}}$$

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HOMOGENEOUS EQUATIONS

■ 1. Find the solution to the differential equation.

$$x^2yy' = x^3 + xy^2$$

Solution:

To put the equation in standard form, we'll divide through by x^2y.

$$\frac{x^2y}{x^2y}y' = \frac{x^3}{x^2y} + \frac{xy^2}{x^2y}$$

$$y' = \frac{x}{y} + \frac{y}{x}$$

Substitute $v = y/x$ and $y' = v + xv'$.

$$v + xv' = v + \frac{1}{v}$$

Now we should have a separable differential equation, so we'll separate variables,

$$xv' = \frac{1}{v}$$

$$x\frac{dv}{dx} = \frac{1}{v}$$

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$$v\,dv = \frac{1}{x}\,dx$$

and then integrate both sides.

$$\int v\,dv = \int \frac{1}{x}\,dx$$

$$\frac{v^2}{2} = \ln x + C$$

$$v^2 = 2\ln x + C$$

$$v = \pm\sqrt{2\ln x + C}$$

Back-substitute for v.

$$\frac{y}{x} = \pm\sqrt{2\ln x + C}$$

$$y = \pm x\sqrt{2\ln x + C}$$

2. Use a substitution to find a solution to the homogeneous equation.

$$x(x+y)y' - y(2x+y) = 0$$

Solution:

Rewrite the equation to solve for y'.

$$x(x+y)y' = y(2x+y)$$

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$$x(x+y)y' = 2xy + y^2$$

$$y' = \frac{2xy + y^2}{x^2 + xy}$$

We'll multiply through the numerator and denominator on the right side by $1/x^2$.

$$y' = \frac{2xy + y^2}{x^2 + xy} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}}$$

$$y' = \frac{2xy\frac{1}{x^2} + y^2\frac{1}{x^2}}{x^2\frac{1}{x^2} + xy\frac{1}{x^2}}$$

$$y' = \frac{2\frac{y}{x} + \frac{y^2}{x^2}}{1 + \frac{y}{x}}$$

Substitute $v = y/x$ and $y' = v + xv'$.

$$v + xv' = \frac{2v + v^2}{1 + v}$$

Now we should have a separable differential equation, so we'll separate variables,

$$xv' = \frac{2v + v^2}{1 + v} - v$$

$$xv' = \frac{2v + v^2 - v - v^2}{1 + v}$$

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$$xv' = \frac{v}{1+v}$$

$$x\frac{dv}{dx} = \frac{v}{1+v}$$

$$\frac{1+v}{v}\,dv = \frac{1}{x}\,dx$$

and then integrate both sides.

$$\int \frac{1+v}{v}\,dv = \int \frac{1}{x}\,dx$$

$$\ln v + v = \ln x + C$$

$$e^{\ln v + v} = e^{\ln x + C}$$

$$ve^{v} = Cx$$

Back-substitute for v.

$$\frac{y}{x}e^{\frac{y}{x}} = Cx$$

$$ye^{\frac{y}{x}} = Cx^2$$

■ 3. Rewrite the homogeneous equation in terms of y/x, but don't solve the separable equation.

$$yy' - x = \sqrt{x^2 + y^2}$$

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Solution:

Solve for y' by dividing both sides by y.

$$yy' = \sqrt{x^2 + y^2} + x$$

$$y' = \frac{\sqrt{x^2 + y^2} + x}{y}$$

Break the fraction in two, then bring the entire first fraction under the root.

$$y' = \frac{\sqrt{x^2 + y^2}}{y} + \frac{x}{y}$$

$$y' = \sqrt{\frac{x^2 + y^2}{y^2}} + \frac{x}{y}$$

Split the fraction under the root in two, simplify, then rewrite the fractions to put them in terms of y/x.

$$y' = \sqrt{\frac{x^2}{y^2} + 1} + \frac{x}{y}$$

$$y' = \sqrt{\frac{1}{\left(\frac{y}{x}\right)^2} + 1} + \frac{1}{\frac{y}{x}}$$

4. Use a substitution to find a solution to the homogeneous equation.

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$$xyy' = x^2 e^{-\frac{y^2}{x^2}} + y^2$$

Solution:

To put the equation in standard form, we'll divide through by xy.

$$\frac{xyy'}{xy} = \frac{x^2 e^{-\frac{y^2}{x^2}}}{xy} + \frac{y^2}{xy}$$

$$y' = \frac{x}{y} e^{-\frac{y^2}{x^2}} + \frac{y}{x}$$

Substitute $v = y/x$ and $y' = v + xv'$.

$$v + xv' = \frac{1}{v} e^{-v^2} + v$$

Now we should have a separable differential equation, so we'll separate variables,

$$xv' = \frac{1}{v} e^{-v^2}$$

$$x\frac{dv}{dx} = \frac{1}{v} e^{-v^2}$$

$$ve^{v^2}\, dv = \frac{1}{x}\, dx$$

and then integrate both sides.

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$$\int ve^{v^2}\,dv = \int \frac{1}{x}\,dx$$

$$\frac{1}{2}e^{v^2} = \ln x + C$$

$$e^{v^2} = 2\ln x + C$$

$$v^2 = \ln|2\ln x + C|$$

$$v = \pm\sqrt{\ln|2\ln x + C|}$$

Back-substitute for v.

$$\frac{y}{x} = \pm\sqrt{\ln|2\ln x + C|}$$

$$y = \pm x\sqrt{\ln|2\ln x + C|}$$

■ 5. Find the solution to the differential equation, if $y(1) = 1$.

$$x^2y' = 3xy + 2y^2$$

Solution:

To put the equation in standard form, we'll divide through by x^2.

$$\frac{x^2y'}{x^2} = \frac{3xy}{x^2} + \frac{2y^2}{x^2}$$

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$$y' = \frac{3y}{x} + \frac{2y^2}{x^2}$$

Substitute $v = y/x$ and $y' = v + xv'$.

$$v + xv' = 3v + 2v^2$$

Now we should have a separable differential equation, so we'll separate variables,

$$xv' = 2v + 2v^2$$

$$x\frac{dv}{dx} = 2v + 2v^2$$

$$\frac{1}{2v + 2v^2}\, dv = \frac{1}{x}\, dx$$

and then integrate both sides and solve for v.

$$\int \frac{1}{2v + 2v^2}\, dv = \int \frac{1}{x}\, dx$$

$$\frac{1}{2}(\ln v - \ln(v + 1)) = \ln x + C$$

$$\ln v - \ln(v + 1) = 2\ln x + C$$

$$\ln \frac{v}{v + 1} = 2\ln x + C$$

$$\ln \frac{v}{v + 1} = \ln x^2 + C$$

$$\frac{v}{v + 1} = Cx^2$$

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$$v = Cx^2v + Cx^2$$

$$v(1 - Cx^2) = Cx^2$$

$$v = \frac{Cx^2}{1 - Cx^2}$$

Back-substitute for v.

$$\frac{y}{x} = \frac{Cx^2}{1 - Cx^2}$$

$$y = \frac{Cx^3}{1 - Cx^2}$$

Once we have this general solution, we recognize from the initial condition $y(1) = 1$ that $x = 1$ and $y = 1$, so we'll plug these values into the general solution to solve for C.

$$1 = \frac{C}{1 - C}$$

$$1 - C = C$$

$$C = \frac{1}{2}$$

Then the solution to the differential equation is

$$y = \frac{\frac{1}{2}x^3}{1 - \frac{1}{2}x^2}$$

$$y = \frac{x^3}{2 - x^2}$$

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■ 6. Use a substitution to find a solution to the homogeneous equation, if $y(1) = 1$.

$$xy' + y\ln x - y = y\ln y$$

Solution:

To put the equation in standard form, we'll divide through by x.

$$xy' = y\ln y - y\ln x + y$$

$$y' = \frac{y}{x}\ln y - \frac{y}{x}\ln x + \frac{y}{x}$$

$$y' = \frac{y}{x}\ln\frac{y}{x} + \frac{y}{x}$$

Substitute $v = y/x$ and $y' = v + xv'$.

$$v + xv' = v\ln v + v$$

Now we should have a separable differential equation, so we'll separate variables,

$$xv' = v\ln v$$

$$x\frac{dv}{dx} = v\ln v$$

$$\frac{1}{v\ln v}\,dv = \frac{1}{x}\,dx$$

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and then integrate both sides.

$$\int \frac{1}{v \ln v} \, dv = \int \frac{1}{x} \, dx$$

$$\ln(\ln v) = \ln x + C$$

$$\ln v = Cx$$

$$v = e^{Cx}$$

Back-substitute for v.

$$\frac{y}{x} = e^{Cx}$$

$$y = xe^{Cx}$$

Once we have this general solution, we recognize from the initial condition $y(1) = 1$ that $x = 1$ and $y = 1$, so we'll plug these values into the general solution to solve for C.

$$1 = e^{C}$$

$$C = 0$$

So the particular solution to the differential equation is

$$y = xe^{0}$$

$$y = x$$

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EXACT EQUATIONS

■ 1. If the differential equation is exact, find its solution.

$$2xy + (x^2 - y^2)\frac{dy}{dx} = 0$$

Solution:

We'll check to see that the equation is exact.

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial}{\partial y}(2xy) = \frac{\partial}{\partial x}(x^2 - y^2)$$

$$2x = 2x$$

The functions $M(x, y)$ and $N(x, y)$ are equally easy to integrate, so we'll just use $M(x, y)$, and then Ψ can be given by

$$\Psi = \int M(x, y)\ dx + h(y)$$

$$\Psi = \int 2xy\ dx + h(y)$$

$$\Psi = x^2y + h(y)$$

We'll differentiate both sides with respect to y,

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$$\Psi_y = x^2 + h'(y)$$

and then substitute $\Psi_y = N(x, y)$ to solve for $h'(y)$.

$$x^2 - y^2 = x^2 + h'(y)$$

$$-y^2 = h'(y)$$

To find $h(y)$, we'll integrate both sides of this equation with respect to y.

$$\int -y^2 \, dy = \int h'(y) \, dy$$

$$-\frac{y^3}{3} + k_1 = h(y) + k_2$$

$$h(y) = -\frac{y^3}{3} + k_1 - k_2$$

$$h(y) = k - \frac{y^3}{3}$$

Plugging this value for $h(y)$ into the equation for Ψ gives

$$\Psi = x^2 y + h(y)$$

$$\Psi = x^2 y + k - \frac{y^3}{3}$$

Finally, setting $\Psi = c_1$ to find the solution to the exact differential equation, we get

$$c_1 = x^2 y + k - \frac{y^3}{3}$$

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$$c_1 - k = x^2y - \frac{y^3}{3}$$

$$c = x^2y - \frac{y^3}{3}$$

■ 2. What is the solution to the exact differential equation?

$$(6y + xe^{-y})\,dy - e^{-y}\,dx = 0$$

Solution:

We'll check to see that the equation is exact.

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial}{\partial y}(-e^{-y}) = \frac{\partial}{\partial x}(6y + xe^{-y})$$

$$e^{-y} = e^{-y}$$

The functions $M(x, y)$ and $N(x, y)$ are equally easy to integrate, so we'll just use $M(x, y)$, and then Ψ can be given by

$$\Psi = \int M(x, y)\,dx + h(y)$$

$$\Psi = \int -e^{-y}\,dx + h(y)$$

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$$\Psi = -xe^{-y} + h(y)$$

We'll differentiate both sides with respect to y,

$$\Psi_y = xe^{-y} + h'(y)$$

and then substitute $\Psi_y = N(x, y)$ to solve for $h'(y)$.

$$6y + xe^{-y} = xe^{-y} + h'(y)$$

$$6y = h'(y)$$

To find $h(y)$, we'll integrate both sides of this equation with respect to y.

$$\int 6y \, dy = \int h'(y) \, dy$$

$$3y^2 + k_1 = h(y) + k_2$$

$$h(y) = 3y^2 + k_1 - k_2$$

$$h(y) = 3y^2 + k$$

Plugging this value for $h(y)$ into the equation for Ψ gives

$$\Psi = -xe^{-y} + h(y)$$

$$\Psi = -xe^{-y} + 3y^2 + k$$

Finally, setting $\Psi = c_1$ to find the solution to the exact differential equation, we get

$$c_1 = -xe^{-y} + 3y^2 + k$$

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$$c_1 - k = -xe^{-y} + 3y^2$$

$$c = -xe^{-y} + 3y^2$$

■ 3. Find the value of a that make the equation an exact differential equation.

$$(5\sin y - x^3)\,dx + ax\cos y\,dy = 0$$

Solution:

Since the equation is exact, we can use our partial derivatives test to find the value of a.

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial}{\partial y}(5\sin y - x^3) = \frac{\partial}{\partial x}(ax\cos y)$$

$$5\cos y = a\cos y$$

These partial derivatives are equal when $a = 5$, so the equation is exact when $a = 5$.

■ 4. If the differential equation is exact, find its solution.

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$$(x \sin y - 2x)\, dx + \left(\frac{x^2}{2} \cos y - 2y \right)\, dy = 0$$

Solution:

We'll check to see that the equation is exact.

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial}{\partial y}(x \sin y - 2x) = \frac{\partial}{\partial x}\left(\frac{x^2}{2} \cos y - 2y \right)$$

$$x \cos y = x \cos y$$

The functions $M(x, y)$ and $N(x, y)$ are equally easy to integrate, so we'll just use $M(x, y)$, and then Ψ can be given by

$$\Psi = \int M(x, y)\, dx + h(y)$$

$$\Psi = \int (x \sin y - 2x)\, dx + h(y)$$

$$\Psi = \frac{x^2}{2} \sin y - x^2 + h(y)$$

We'll differentiate both sides with respect to y,

$$\Psi_y = \frac{x^2}{2} \cos y + h'(y)$$

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and then substitute $\Psi_y = N(x, y)$ to solve for $h'(y)$.

$$\frac{x^2}{2}\cos y - 2y = \frac{x^2}{2}\cos y + h'(y)$$

$$-2y = h'(y)$$

To find $h(y)$, we'll integrate both sides of this equation with respect to y.

$$\int -2y \, dy = \int h'(y) \, dy$$

$$-y^2 + k_1 = h(y) + k_2$$

$$h(y) = -y^2 + k_1 - k_2$$

$$h(y) = -y^2 + k$$

Plugging this value for $h(y)$ into the equation for Ψ gives

$$\Psi = \frac{x^2}{2}\sin y - x^2 + h(y)$$

$$\Psi = \frac{x^2}{2}\sin y - x^2 - y^2 + k$$

Finally, setting $\Psi = c_1$ to find the solution to the exact differential equation, we get

$$c_1 = \frac{x^2}{2}\sin y - x^2 - y^2 + k$$

$$c_1 - k = \frac{x^2}{2}\sin y - x^2 - y^2$$

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$$c = \frac{x^2}{2}\sin y - x^2 - y^2$$

■ 5. Find the value of a that makes the equation an exact differential equation.

$$(2\cos x + 4\ln y)\,dx + \left(\frac{a^2 x}{y} + e^y\right)\,dy = 0$$

Solution:

Since the equation is exact, we can use our partial derivatives test to find the value of a.

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial}{\partial y}(2\cos x + 4\ln y) = \frac{\partial}{\partial x}\left(\frac{a^2 x}{y} + e^y\right)$$

$$\frac{4}{y} = \frac{a^2}{y}$$

$$a^2 = 4$$

$$a = \pm 2$$

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The partial derivatives are equal when $a = \pm 2$, which means the equation is exact when $a = \pm 2$.

■ 6. What is the solution to the exact differential equation?

$$(x \cos y + e^y)\, dy + (e^x + \sin y)\, dx = 0$$

Solution:

We'll check to see that the equation is exact.

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\frac{\partial}{\partial y}(e^x + \sin y) = \frac{\partial}{\partial x}(x \cos y + e^y)$$

$$\cos y = \cos y$$

The functions $M(x, y)$ and $N(x, y)$ are equally easy to integrate, so we'll just use $M(x, y)$, and then Ψ can be given by

$$\Psi = \int M(x, y)\, dx + h(y)$$

$$\Psi = \int (e^x + \sin y)\, dx + h(y)$$

$$\Psi = e^x + x \sin y + h(y)$$

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We'll differentiate both sides with respect to y,

$$\Psi_y = x\cos y + h'(y)$$

and then substitute $\Psi_y = N(x, y)$ to solve for $h'(y)$.

$$x\cos y + e^y = x\cos y + h'(y)$$

$$e^y = h'(y)$$

To find $h(y)$, we'll integrate both sides of this equation with respect to y.

$$\int e^y \, dy = \int h'(y) \, dy$$

$$e^y + k_1 = h(y) + k_2$$

$$h(y) = e^y + k_1 - k_2$$

$$h(y) = e^y + k$$

Plugging this value for $h(y)$ into the equation for Ψ gives

$$\Psi = e^x + x\sin y + h(y)$$

$$\Psi = e^x + x\sin y + e^y + k$$

Finally, setting $\Psi = c_1$ to find the solution to the exact differential equation, we get

$$c_1 = e^x + x\sin y + e^y + k$$

$$c_1 - k = e^x + x\sin y + e^y$$

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$$c = e^x + x \sin y + e^y$$

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